Additional file 1: Supplementary tables, figure, and analysis plan amendments

Separating definition from denominator in NHANES diabetes prevalence estimates: a cross-sectional, design-based comparison of HbA1c and fasting glucose surveillance definitions

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All tables give the sample universe, weight, and unweighted denominator. Percentages are weighted estimates with 95% Korn-Graubard intervals unless stated otherwise. Estimates that do not meet the NCHS Data Presentation Standards for Proportions are shown as "suppressed"; their values are kept in the aggregate files, marked as suppressed, for audit only. Run identifier: 20261007T212439Z.

Table S1. Configuration checker refusals

These requests were submitted to the checker on purpose for each release, and each was refused with the reason shown. The 2017 to March 2020 release has no phlebotomy weight, so its examination weight WTMECPRP is the documented HbA1c weight and is accepted in domain H; the corresponding August 2021 to August 2023 request with WTMEC2YR is refused because WTPH2YR is the documented weight there.

Release	Request	Checker result	Reason given
Aug 2021-Aug 2023	FPG126 in domain H with WTPH2YR	refused	FPG126 needs fpg but domain H does not guarantee it. The phenotype needs a measure that this domain does not guarantee, so its denominator would silently shrink. Fasting glucose must be analyzed with the fasting subsample weight in a fasting domain (F, P, or P_NARROW).
Aug 2021-Aug 2023	A1C65 in domain H with WTMEC2YR	refused	WTMEC2YR (exam) used with domain H. This weight does not belong to the domain's sample. Use the weight listed for the domain. Allowed: WTPH2YR.
Aug 2021-Aug 2023	FPG126 in domain F with WTMEC2YR	refused	Fasting glucose must be analyzed with the fasting subsample weight in a fasting domain (F, P, or P_NARROW). WTMEC2YR (exam) used with domain F. This weight does not belong to the domain's sample. Use the weight listed for the domain. Allowed: WTSAF2YR.
Aug 2021-Aug 2023	TOTAL_A1C_FPG in domain H with WTPH2YR	refused	TOTAL_A1C_FPG needs fpg but domain H does not guarantee it. The phenotype needs a measure that this domain does not guarantee, so its denominator would silently shrink. Fasting glucose must be analyzed with the fasting subsample weight in a fasting domain (F, P, or P_NARROW).
Aug 2021-Aug 2023	A1C65 in domain P with WTSAF2YR (imputed glucose)	refused	Fasting glucose imputed from HbA1c is not an independent measurement and is not supported.
Aug 2021-Aug 2023	A1C65 in domain P with WTSAF2YR (pooled releases)	refused	Releases are analyzed separately. Pooling 2017-March 2020 with August 2021-August 2023 by dividing weights is not supported.
Aug 2021-Aug 2023	K3 vs K2 labeled as a definition difference	refused	Two different phenotypes estimated in different domains or with different weights cannot be compared as a definition difference. Estimate both in domain P with the fasting weight.
2017-Mar 2020	FPG126 in domain H with WTMECPRP	refused	FPG126 needs fpg but domain H does not guarantee it. The phenotype needs a measure that this domain does not guarantee, so its denominator would silently shrink. Fasting glucose must be analyzed with the fasting subsample

Release	Request	Checker result	Reason given
			weight in a fasting domain (F, P, or P_NARROW).
2017-Mar 2020	FPG126 in domain F with WTMECPRP	refused	Fasting glucose must be analyzed with the fasting subsample weight in a fasting domain (F, P, or P_NARROW). WTMECPRP (exam) used with domain F. This weight does not belong to the domain's sample. Use the weight listed for the domain. Allowed: WTSAFPRP.
2017-Mar 2020	TOTAL_A1C_FPG in domain H with WTMECPRP	refused	TOTAL_A1C_FPG needs fpg but domain H does not guarantee it. The phenotype needs a measure that this domain does not guarantee, so its denominator would silently shrink. Fasting glucose must be analyzed with the fasting subsample weight in a fasting domain (F, P, or P_NARROW).
2017-Mar 2020	A1C65 in domain P with WTSAFPRP (imputed glucose)	refused	Fasting glucose imputed from HbA1c is not an independent measurement and is not supported.
2017-Mar 2020	A1C65 in domain P with WTSAFPRP (pooled releases)	refused	Releases are analyzed separately. Pooling 2017-March 2020 with August 2021-August 2023 by dividing weights is not supported.
2017-Mar 2020	K3 vs K2 labeled as a definition difference	refused	Two different phenotypes estimated in different domains or with different weights cannot be compared as a definition difference. Estimate both in domain P with the fasting weight.

Table S2. Automated checks

Software checks test the code against itself and a second implementation. Source reconciliation compares the data files with the NCHS codebooks. External reconciliation compares estimates with published NCHS aggregates. None of these is independent expert review.

Release	Kind	Check	Result	Detail
2021-2023	software check	join cardinality: unique SEQN, every component SEQN in DEMO	pass	every component has unique SEQN and every SEQN is in DEMO
2021-2023	software check	analytic frame rows equal DEMO rows	pass	11933 rows
2021-2023	software check	fasting weight outside window	pass	count = 0
2021-2023	software check	domain nesting violations	pass	count = 0
2021-2023	software check	null phenotype inside domain	pass	count = 0
2021-2023	software check	laboratory weight without examination	pass	count = 0
2021-2023	source reconciliation	file counts equal published codebook frequencies	pass	91 of 91 codebook rows match
2021-2023	source reconciliation	glucose values with zero fasting weight equal the documented count	pass	311 in the file, 311 in the documentation; window breakdown <8 h, >=24 h, unknown, inside: [277, 32, 2, 0]
2021-2023	software check	all configurations and contrasts accepted by the checker	pass	8 configurations plus 3 auxiliary or sensitivity estimates, 12 contrasts
2021-2023	software check	invalid requests refused by the checker	pass	7 refusal examples
2021-2023	software check	direct weighted proportions equal R survey estimates	pass	max abs difference 3.75e-16 over 23 estimates and contrasts

Release	Kind	Check	Result	Detail
2021-2023	software check	second implementation of linearized SE equals R survey SE	pass	max relative difference 1.83e-12
2021-2023	software check	direct weighted 2x2 cross-tab equals R cell estimates	pass	max abs difference 2.22e-16
2021-2023	software check	second implementation reproduces every other reported table	pass	162 point estimates (agreement, kappa and ratios, overlap, sensitivity 3, missingness, groups, post hoc, reconciliation), max abs difference 1.67e-15; 143 standard errors, max relative difference 4.80e-14; kappa and ratio standard errors not recomputed
2021-2023	software check	decompositions add up exactly	pass	max error 8.3e-17
2021-2023	software check	paired SE equals joint-totals SE for same-domain contrasts	pass	max difference 4.3e-18
2021-2023	software check	no singleton strata in the full design or analysis domains	pass	design df 15; domain df {'A': 15, 'H': 15, 'F': 15, 'P': 15, 'P_NARROW': 15}
2021-2023	external reconciliation	sample sizes equal NCHS Data Brief 516	pass	18 of 18
2021-2023	external reconciliation	percents round to NCHS Data Brief 516 values	pass	18 of 18
2021-2023	external reconciliation	standard errors round to NCHS Data Brief 516 values	pass	18 of 18
2021-2023	external reconciliation	Korn-Graubard intervals reconcile with NCHS Data Brief 516	pass	17 of 18 match with the effective sample size capped at n as in the NCHS standard; 18 of 18 match without the cap
2017-2020	software check	join cardinality: unique SEQN, every component SEQN in DEMO	pass	every component has unique SEQN and every SEQN is in DEMO
2017-2020	software check	analytic frame rows equal DEMO rows	pass	15560 rows
2017-2020	software check	fasting weight outside window	pass	count = 0
2017-2020	software check	domain nesting violations	pass	count = 0
2017-2020	software check	null phenotype inside domain	pass	count = 0
2017-2020	software check	laboratory weight without examination	pass	count = 0
2017-2020	source reconciliation	file counts equal published codebook frequencies	pass	85 of 85 codebook rows match
2017-2020	source reconciliation	glucose values with zero fasting weight equal the documented count	pass	268 in the file, 268 in the documentation; window breakdown <8 h, >=24 h, unknown, inside: [225, 43, 0, 0]
2017-2020	software check	all configurations and contrasts accepted by the checker	pass	8 configurations plus 3 auxiliary or sensitivity estimates, 12 contrasts
2017-2020	software check	invalid requests refused by the checker	pass	6 refusal examples
2017-2020	software check	direct weighted proportions equal R survey estimates	pass	max abs difference 6.73e-16 over 23 estimates and contrasts
2017-2020	software check	second implementation of linearized SE equals R survey SE	pass	max relative difference 7.07e-13
2017-2020	software check	direct weighted 2x2 cross-tab equals R cell estimates	pass	max abs difference 9.99e-16

Release	Kind	Check	Result	Detail
2017-2020	software check	second implementation reproduces every other reported table	pass	143 point estimates (agreement, kappa and ratios, overlap, sensitivity 3, missingness, groups, post hoc, reconciliation), max abs difference 3.89e-15; 124 standard errors, max relative difference 4.60e-14; kappa and ratio standard errors not recomputed
2017-2020	software check	decompositions add up exactly	pass	max error 4.2e-16
2017-2020	software check	paired SE equals joint-totals SE for same-domain contrasts	pass	max difference 7.8e-18
2017-2020	software check	no singleton strata in the full design or analysis domains	pass	design df 25; domain df {'A': 25, 'H': 25, 'F': 25, 'P': 25, 'P_NARROW': 25}
2017-2020	external reconciliation	sample sizes equal NCHS Data Brief 516	pass	1 of 1

Table S3. Participant flow

Each step is applied after the steps above it in the same branch. The HbA1c and fasting branches both start from domain A. Counts are unweighted; the last column names the weight carried at that step.

Release	Step	Branch	Description	Remaining n (unweighted)	Removed n	Weight carried at this step
Aug 2021-Aug 2023	1	all	Participants in the demographics file	11,933		interview weight
Aug 2021-Aug 2023	2	all	Interviewed and examined (RIDSTATR = 2)	8,860	3,073	examination weight
Aug 2021-Aug 2023	3	all	Aged 20 or older	6,064	2,796	examination weight
Aug 2021-Aug 2023	4	all	Excluding pregnant at the examination (RIDEXPRG = 1)	6,023	41	examination weight
Aug 2021-Aug 2023	5	all	Excluding pregnancy not ascertained at the examination but reported at the interview (RIDEXPRG = 3 and RHD143 = 1)	6,022	1	examination weight
Aug 2021-Aug 2023	6	all	Excluding DIQ010 refused, don't know, or missing: domain A	6,022	0	examination weight
Aug 2021-Aug 2023	7	hba1c	Excluding zero phlebotomy weight (no blood specimen) or absent from the HbA1c file	5,754	268	phlebotomy weight (WTPH2YR)
Aug 2021-Aug 2023	8	hba1c	Excluding positive weight but no HbA1c result: domain H	5,729	25	phlebotomy weight (WTPH2YR)
Aug 2021-Aug 2023	9	fasting	Assigned to the fasting subsample (record in GLU file)	3,396	2,626	examination weight

Release	Step	Branch	Description	Remaining n (unweighted)	Removed n	Weight carried at this step
Aug 2021-Aug 2023	10	fasting	Excluding subsample members with no blood specimen	3,185	211	examination weight
Aug 2021-Aug 2023	11	fasting	Excluding fast outside 8 to less than 24 hours (zero fasting weight, glucose not used)	2,940	245	fasting subsample weight
Aug 2021-Aug 2023	12	fasting	Excluding positive weight but no glucose result: domain F	2,940	0	fasting subsample weight
Aug 2021-Aug 2023	13	paired	Excluding no HbA1c result: common paired domain P	2,938	2	fasting subsample weight
Aug 2021-Aug 2023	14	paired	Excluding intake within 8 hours or unknown on any FASTQX item: domain P-narrow	2,766	172	fasting subsample weight
2017-Mar 2020	1	all	Participants in the demographics file	15,560		interview weight
2017-Mar 2020	2	all	Interviewed and examined (RIDSTATR = 2)	14,300	1,260	examination weight
2017-Mar 2020	3	all	Aged 20 or older	8,544	5,756	examination weight
2017-Mar 2020	4	all	Excluding pregnant at the examination (RIDEXPRG = 1)	8,457	87	examination weight
2017-Mar 2020	5	all	Excluding pregnancy not ascertained at the examination but reported at the interview (RIDEXPRG = 3 and RHD143 = 1)	8,456	1	examination weight
2017-Mar 2020	6	all	Excluding DIQ010 refused, don't know, or missing: domain A	8,451	5	examination weight
2017-Mar 2020	7	hba1c	Record in the HbA1c file (no phlebotomy weight in this release, so no adult is removed here)	8,451	0	examination weight (WTMECPRP); no phlebotomy weight in this release
2017-Mar 2020	8	hba1c	Excluding no HbA1c result, including adults who gave no blood specimen: domain H	8,000	451	examination weight (WTMECPRP); no phlebotomy weight in this release
2017-Mar 2020	9	fasting	Assigned to the fasting subsample (record in GLU file)	4,210	4,241	examination weight

Release	Step	Branch	Description	Remaining n (unweighted)	Removed n	Weight carried at this step
2017-Mar 2020	10	fasting	Excluding subsample members with no blood specimen	3,965	245	examination weight
2017-Mar 2020	11	fasting	Excluding fast outside 8 to less than 24 hours (zero fasting weight, glucose not used)	3,767	198	fasting subsample weight
2017-Mar 2020	12	fasting	Excluding positive weight but no glucose result: domain F	3,767	0	fasting subsample weight
2017-Mar 2020	13	paired	Excluding no HbA1c result: common paired domain P	3,757	10	fasting subsample weight
2017-Mar 2020	14	paired	Excluding intake within 8 hours or unknown on any FASTQX item: domain P-narrow	3,591	166	fasting subsample weight

Table S4. Laboratory status by group among adults in domain A, August 2021 to August 2023

Weighted with the examination weight (WTMEC2YR). “Not assigned” means the participant was not selected for the morning fasting subsample.

Group (domain A, WTMEC2YR)	Denominator n	Valid fasting glucose %	Not assigned %	Assigned, no blood %	Fast outside window %	No HbA1c %
All adults in domain A	6,022	48.9	43.7	3.5	3.9	4.9
20-44	1,964	47.1	44.7	4.1	4.1	5.7
45-64	2,052	50.1	43.6	2.8	3.5	3.8
65+	2,006	50.5	41.8	3.4	4.2	4.8
Female	3,302	50.1	43.2	3.4	3.3	4.7
Male	2,720	47.5	44.3	3.6	4.6	5.1
Mexican American	401	53.3	40.6	suppressed	2.7	4.4
Non-Hispanic Asian	335	50.3	41.0	suppressed	suppressed	suppressed
Non-Hispanic Black	743	41.9	45.9	5.4	6.8	8.3
Non-Hispanic White	3,549	50.1	43.4	3.1	3.4	4.0
Other Hispanic	608	47.3	43.7	3.3	suppressed	4.5
Other or multiracial	386	44.8	49.1	4.1	2.0	7.3

Table S5. Reconciliation with NCHS Data Brief 516

Published values are copied from NCHS Data Brief 516 (adults 20 and older, fasting sample weights). My values use domain P with WTSAF2YR. A match means my value rounds to the published value. The one interval that differs matches when the effective sample size is not capped at the sample size.

NCHS table	Measure	Group	n, published / study	Percent, published / study	95% CI, published / study	SE, published / study	CI matches
Table 1	total	all	2,938 / 2,938	15.8 / 15.79	13.6 to 18.2 / 13.59 to 18.19	1.1 / 1.06	yes

NCHS table	Measure	Group	n, published / study	Percent, published / study	95% CI, published / study	SE, published / study	CI matches
Table 1	total	men	1,306 / 1,306	18.0 / 17.97	15.7 to 20.4 / 15.69 to 20.43	1.1 / 1.09	yes
Table 1	total	women	1,632 / 1,632	13.7 / 13.66	11.0 to 16.7 / 11.00 to 16.69	1.3 / 1.30	yes
Table 1	diagnosed	all	2,938 / 2,938	11.3 / 11.30	9.3 to 13.5 / 9.31 to 13.55	1.0 / 0.97	yes
Table 1	diagnosed	men	1,306 / 1,306	12.9 / 12.90	11.0 to 15.0 / 10.98 to 15.02	0.9 / 0.91	yes
Table 1	diagnosed	women	1,632 / 1,632	9.7 / 9.73	7.3 to 12.7 / 7.28 to 12.68	1.2 / 1.22	yes
Table 1	undiagnosed	all	2,938 / 2,938	4.5 / 4.49	3.3 to 5.9 / 3.34 to 5.88	0.6 / 0.57	yes
Table 1	undiagnosed	men	1,306 / 1,306	5.1 / 5.07	3.6 to 6.9 / 3.57 to 6.95	0.8 / 0.76	yes
Table 1	undiagnosed	women	1,632 / 1,632	3.9 / 3.92	2.8 to 5.3 / 2.83 to 5.29	0.6 / 0.55	yes
Table 2	total	20-39	688 / 688	3.6 / 3.57	2.0 to 5.9 / 1.95 to 5.93	0.9 / 0.87	yes
Table 2	total	40-59	861 / 861	17.7 / 17.70	14.7 to 21.0 / 14.74 to 20.99	1.4 / 1.43	yes
Table 2	total	60+	1,389 / 1,389	27.3 / 27.26	23.0 to 31.9 / 22.96 to 31.89	2.0 / 2.05	yes
Table 2	diagnosed	20-39	688 / 688	2.2 / 2.24	1.1 to 4.0 / 1.10 to 4.03	0.6 / 0.63	yes
Table 2	diagnosed	40-59	861 / 861	12.1 / 12.10	10.0 to 14.5 / 9.83 to 14.69	1.0 / 1.04	no (yes without the cap)
Table 2	diagnosed	60+	1,389 / 1,389	20.5 / 20.47	16.2 to 25.3 / 16.20 to 25.29	2.1 / 2.07	yes
Table 2	undiagnosed	20-39	688 / 688	1.3 / 1.33	0.5 to 3.0 / 0.45 to 3.00	0.5 / 0.53	yes
Table 2	undiagnosed	40-59	861 / 861	5.6 / 5.60	3.3 to 8.9 / 3.25 to 8.90	1.2 / 1.24	yes
Table 2	undiagnosed	60+	1,389 / 1,389	6.8 / 6.79	5.3 to 8.5 / 5.32 to 8.51	0.7 / 0.72	yes

Table S6. Paired HbA1c and glucose estimates by group, August 2021 to August 2023

Component percentages: 95% Korn-Graubard intervals. Differences: 95% t intervals with the degrees of freedom shown. † One component estimate is suppressed; the difference is shown for completeness and is not interpreted.

Group (domain P, WTSAF2YR)	Denominator n	HbA1c ≥ 6.5%, % (95% CI)	Glucose ≥ 126 mg/dL, % (95% CI)	Difference, points (95% CI)	df
20-44	918	3.9 (2.4 to 5.9)	4.3 (2.5 to 6.7)	−0.4 (−1.3 to +0.5)	15
45-64	1,026	15.4 (13.1 to 18.0)	15.4 (13.0 to 18.0)	0.0 (−1.5 to +1.5)	15
65+	994	17.6 (14.1 to 21.5)	18.7 (14.6 to 23.3)	−1.1 (−3.5 to +1.3)	15
Female	1,632	8.7 (6.7 to 11.0)	9.0 (6.9 to 11.5)	−0.3 (−1.6 to +1.0)	15
Male	1,306	12.9 (10.8 to 15.3)	13.5 (11.2 to 16.0)	−0.5 (−1.7 to +0.6)	15
Mexican American	218	15.6 (10.6 to 21.9)	15.7 (10.6 to 21.9)	0.0 (−2.2 to +2.1)	11
Non-Hispanic Asian	166	suppressed	10.0 (5.5 to 16.4)	−2.9 (−7.6 to +1.8) †	11
Non-Hispanic Black	296	19.5 (11.8 to 29.3)	suppressed	+3.3 (−0.4 to +6.9) †	14
Non-Hispanic White	1,791	8.9 (6.8 to 11.3)	9.5 (7.4 to 12.0)	−0.7 (−1.8 to +0.5)	15
Other Hispanic	286	10.0 (6.4 to 14.5)	11.4 (7.2 to 16.8)	−1.4 (−4.6 to +1.7)	12
Other or multiracial	181	13.2 (8.2 to 19.6)	13.9 (8.8 to 20.4)	−0.7 (−3.1 to +1.7)	14

Table S7. Analyses not prespecified

Added after the planned analyses (amendment 8) and exploratory. Domain P with WTSAF2YR; the PH3 rows are restricted to adults in P without a reported diagnosis. PH1 and PH2 are shares of all adults in P, and their sum equals the primary contrast C1. PH4 and PH5 repeat C1 with the HbA1c cut point at 6.4% and 6.6%. Left and right components are weighted percentages with Korn-Graubard intervals; differences, kappa, agreement, and the one-test share use 95% t intervals.

Release	ID	Quantity	Denominator n	Left / right, %	Estimate (95% CI)
Aug 2021-Aug 2023	PH1	No reported diagnosis and HbA1c ≥ 6.5 minus no reported diagnosis and glucose ≥ 126 (share of all adults in P)	2,938	2.6 / 3.7	−1.12 (−1.69 to −0.56)
Aug 2021-Aug 2023	PH2	Reported diagnosis and HbA1c ≥ 6.5 minus reported diagnosis and glucose ≥ 126 (share of all adults in P)	2,938	8.2 / 7.5	+0.71 (+0.20 to +1.22)
Aug 2021-Aug 2023	PH3 kappa	Cohen's kappa, HbA1c ≥ 6.5 versus glucose ≥ 126, adults without a reported diagnosis	2,529		0.55 (0.49 to 0.61)
Aug 2021-Aug 2023	PH3 agreement	Percent agreement, adults without a reported diagnosis	2,529		96.9% (96.0 to 97.8)
Aug 2021-Aug 2023	PH3 one test only	Share above only one threshold among adults above either, no reported diagnosis	2,529		60.7% (55.3 to 66.0)
Aug 2021-Aug 2023	PH4	HbA1c ≥ 6.4 minus glucose ≥ 126	2,938	12.0 / 11.2	+0.85 (−0.10 to +1.80)
Aug 2021-Aug 2023	PH5	HbA1c ≥ 6.6 minus glucose ≥ 126	2,938	9.9 / 11.2	−1.30 (−2.01 to −0.59)
2017-Mar 2020	PH1	No reported diagnosis and HbA1c ≥ 6.5 minus no reported diagnosis and glucose ≥ 126 (share of all adults in P)	3,757	2.4 / 3.8	−1.48 (−2.18 to −0.78)
2017-Mar 2020	PH2	Reported diagnosis and HbA1c ≥ 6.5 minus reported diagnosis and glucose ≥ 126 (share of all adults in P)	3,757	7.9 / 8.0	−0.07 (−1.10 to +0.96)
2017-Mar 2020	PH3 kappa	Cohen's kappa, HbA1c ≥ 6.5 versus glucose ≥ 126, adults without a reported diagnosis	3,137		0.52 (0.44 to 0.60)
2017-Mar 2020	PH3 agreement	Percent agreement, adults without a reported diagnosis	3,137		96.7% (96.0 to 97.5)
2017-Mar 2020	PH3 one test only	Share above only one threshold among adults above either, no reported diagnosis	3,137		63.3% (55.8 to 70.7)
2017-Mar 2020	PH4	HbA1c ≥ 6.4 minus glucose ≥ 126	3,757	11.5 / 11.8	−0.33 (−1.51 to +0.85)
2017-Mar 2020	PH5	HbA1c ≥ 6.6 minus glucose ≥ 126	3,757	9.5 / 11.8	−2.35 (−3.61 to −1.10)

Figure S1

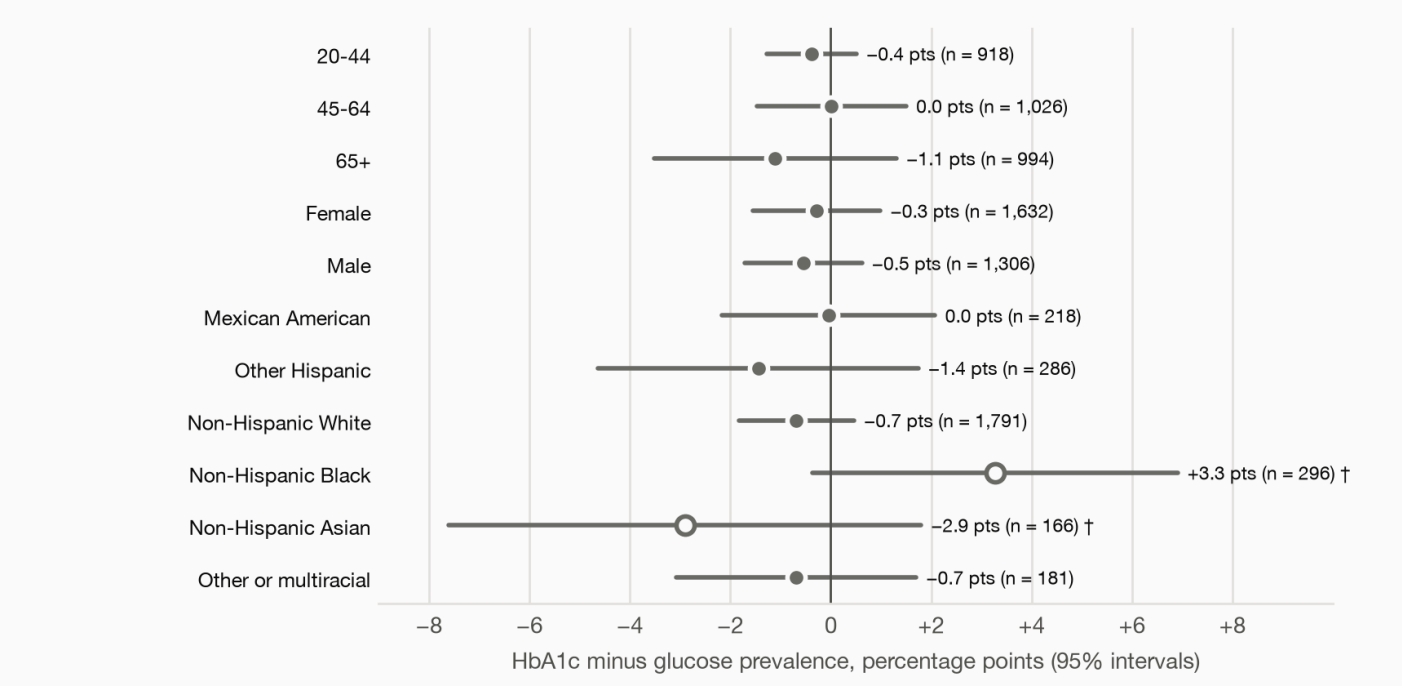


Figure S1. Paired HbA1c minus glucose difference by age, sex, and race and Hispanic origin. Domain P, August 2021 to August 2023, WTSAF2YR, with 95% t intervals. Hollow markers and †: one component estimate is suppressed under the NCHS standards, so the difference is shown for completeness and not interpreted. Race and Hispanic origin is a self-reported social classification.

Analysis plan amendments

All amendments were made on 2026-10-07 after the plan was frozen (09:33Z) and after the first estimation run. No amendment changed a phenotype definition, a threshold, a domain, a 2021-2023 weight, or one of the eight configurations; amendment 2 specified the 2017-March 2020 weights, which the plan left to that release's documentation. The frozen text is preserved unchanged in docs/analysis_plan_frozen_v1.md with its SHA-256 in docs/analysis_plan_frozen_v1.sha256.

1. Contrasts C11 and C12 (HbA1c and glucose each in the narrow fasting domain minus the paired domain) were added to config/analysis_configs.yaml after the first estimation run. They implement the comparison of the narrow and paired domains that sensitivity analysis 1 already describes.
2. The weight rules were rewritten so that each release maps its own weight variables to roles. This was needed to run the planned 2017-March 2020 replication, where no phlebotomy weight exists; HbA1c there uses the examination weight WTMECPRP, following the P_GHB documentation's instruction to use the sample weights in the demographics file.
3. The planned external reconciliation was implemented against NCHS Data Brief 516, Tables 1 and 2 (crude estimates overall, by sex, and by age 20-39, 40-59, and 60 and older). For this comparison only, undiagnosed diabetes was computed as a direct proportion (no reported diagnosis and either biomarker at or above threshold) and the brief's age groups were used. An interval variant without the cap on the effective sample size is computed only to explain one interval that differed from the publication; every reported interval uses the cap required by the NCHS standards. The plan named svyciprop(method = "beta") for Korn-Graubard intervals; every reported interval instead caps the effective sample size at the actual sample size, as the NCHS standards specify. This changes one presentation decision: in sensitivity analysis 3 for 2021-2023, the share of untreated diagnosed adults with both values below threshold (n = 53) is suppressed with the cap (interval width 0.32) and would have been presented without it (width 0.27).
4. Paired differences by group carry a note when one of the two component estimates is suppressed under the NCHS standards.
5. Sensitivity analysis 3 by treatment status reports a "treatment unknown" row (one adult in 2021-2023), which is suppressed by the sample size rule.
6. The 2017-March 2020 replication found that 451 of 8,451 adults in domain A (5.3% as an unweighted count share; 4.2% weighted with WTMECPRP) lacked an HbA1c result under an examination weight that does not adjust for that nonresponse. The plan's 5 percent of a domain rule was applied to the unweighted count share, so this is reported as a limitation and was not reweighted.
7. The prior-work search log (docs/literature_search_log.md) and docs/contribution_decision.md were finalized after the first estimation run, contrary to the planned order of work. The question, estimand, comparator, and configurations had already been fixed in the frozen plan, and no search or screening decision used an estimate.
8. After the planned analyses were complete, three analyses that were not prespecified were added and are reported as exploratory (results/tables/<release>/posthoc.csv; Additional file 1, Table S7): (a) the primary contrast split into its parts among adults without and with a reported diagnosis, each as a share of all adults in P, which sum to C1; (b) agreement (kappa, percent agreement, and the share above only one threshold) among adults in P without a reported diagnosis, the group studied by Carson and colleagues; and (c) the primary contrast with the HbA1c cut point at 6.4% and 6.6%, one reporting unit either side of 6.5%, to show sensitivity to small calibration differences such as the documented HbA1c instrument change. All three use domain P, WTSAF2YR, and the paired estimation of the primary contrast. They do not change any planned result.